

Практикум B2

113 marks, 40 blocks, nine techniques. Two ideas that are one line of algebra each — and the exam pays for neither of them. It pays for the order in which you compose, and for the domain on which the inverse is allowed to live.

О практикуме

Первый практикум серии на английском. Ноутбук, карточка приёмов и сообщения проверок — английские; документация репозитория, включая эту заметку, остаётся русской. Переключает язык сам ноутбук вызовом `language('en')` в установочной ячейке, значение по умолчанию в `kit.py` русское, поэтому десять готовых практикумов и тренажёр не задеты.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-b2-composition-inverse

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Practicum B2: composition and inverse functions

113 marks, 40 blocks, nine techniques. Two ideas that are one line of algebra each — and the exam pays for neither of them. It pays for the **order** in which you compose, and for the **domain** on which the inverse is allowed to live.

Material. 40 blocks and 113 marks from the AA HL archive, sessions May 2021 — May 2025. Five of those marks are one question counted twice: the whole November 2023 session sits in the corpus as two zones, and the papers are the same paper. Every task below is a real past-paper question, with the source given.

This practicum is in English. The first ten were in Russian; the checks now speak whichever language the notebook asks them to, and this one asks for English in the setup cell below.

The main idea. Composition chains two functions; the inverse is the chain run backwards. Written down, both are short. The marks are somewhere else.

- **Order.** $(f \circ g)(x)$ means g first. The M1 is for composing in the right order, and roughly half the composite questions in the archive are built so that the wrong order gives a different, plausible answer.
- **Domain.** An inverse exists only where the function is one-to-one. So the question first asks for the largest a that keeps it so; then domain and range **swap**, so the domain of f^{-1} is the range of f ; and if the algebra ends in a square root, one branch is the inverse and the other is a different function. That last step is an R1 — a sentence, not a formula — and it is the mark most often lost.

How to work

1. Read the map of techniques first. Rungs 1-5 run forwards, rungs 6-9 run

backwards, and the two halves meet at the end.

2. Work **on paper**. Enter exact answers: `sqrt(2)`, `Rational(3, 2)`, `1 + 3*sqrt(2)/2`. Round only where the question says "3 s.f.".
3. Domains and ranges are entered as sets or inequalities: `Interval(0, sqrt(3))`, `(x > 0)`, `Interval.Lopen(Rational(1,2), 1)`. The notation is free; the endpoints are not.
4. An inverse is checked by **undoing**, not against a stored answer. The check substitutes f into what you wrote and asks for x back.
5. The last two blocks are a recognition trainer and one question on a timer.

Difficulty marks:  the technique on its own ·  the technique in a wrapper ·  several techniques, or a whole exam question.

```

import sys
sys.path.append('.')          # from
practicum/functions to practicum/kit.py
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
solveset, Interval, real_roots

language('en')                # this
notebook is in English, and so are the checks

a, b, c, m, p, q = symbols('a b c m p q')
n, r, t = symbols('n r t')

print('ready; sympy', sp.__version__)
print('compose:              ', (lambda f, g:
f.subs(x, g))(x**2 + 1, 3*x - 2))
print('inverse by hand:      ', solve(Eq(y,
(7*x + 7)/(2*x - 4)), x))
print('a domain as a set:    ',
Interval.Lopen(Rational(1, 2), 1))
print('sin of an arcsin:     ',
simplify(sin(asin((x**2 - 1)/(x**2 + 1)))))

```

Map of techniques

#	Technique	Trigger in the question	First move
1	Evaluate a composite	«find $(f \circ g)(a)$ », «find $h(4)$ where $h(x) = f(g(x))$ »	evaluate the inner function at the number
2	Build the composite	«write down an expression for $(g \circ f)(x)$ », «show that $(f \circ g)(x) = \dots$ »	substitute the whole inner expression, in brackets
3	The composite as an equation	«given $(g \circ f)(2) = 10$, find k », «find the two possible f »	build the composite, then set it equal
4	Substitute $-x$	«show that f is an even function»	write $f(-x)$ in full and simplify
5	Compose f with itself	« $f^n(x)$ », «self-composite», «suggest an expression»	compute f^2 and f^3 , keep the pattern visible
6	The defining property	«find $f^{-1}(8)$ », «solve $f^{-1}(2x - 7) = -3$ », « $g = g^{-1}$ »	turn it around: $f^{-1}(a) = b$ is $f(b) = a$
7	Swap and solve	«find an expression for	write $y = f(x)$, interchange x

#	Technique	Trigger in the question	First move
		$f^{-1}(x)$ »	and y
8	Existence and domain	«largest a for which f^{-1} exists», «state the domain of f^{-1} »	find where f turns; then range of $f =$ domain of f^{-1}
9	Choosing the branch	$a \pm$ appears, and «justifying your answer»	range of $f^{-1} =$ domain of f — that is what chooses

The ladder runs in two directions. Rungs 1–3 go **forwards**: evaluate a composite, write one down, then set it equal to something and solve. Rung 4 is the smallest composite there is, $f(-x)$, and rung 5 composes f with itself until a pattern appears.

Rungs 6–9 go **backwards**. First the defining property, which answers most inverse questions without ever finding the inverse. Then the swap that produces the formula. Then the domain the formula is allowed to live on. Then the

choice of branch, which is where the R1 sits.

The loop closes at the end. A function is its own inverse exactly when $f \circ f$ is the identity, and exactly when its graph is symmetric in $y = x$. Task 13 finds such a function by algebra; the timed task meets the same fact again as the case $m = -1$ of a Paper 3 investigation.

What saves the most time. Before composing, look at which function is written next to the bracket. Before finding an inverse, look at whether the question actually wants the formula — half the time it wants one value, and $f^{-1}(a) = b \iff f(b) = a$ gets it in one line.

Theory 1. Composition, and why the order is the mark

$(f \circ g)(x)$ means $f(g(x))$: **the function written next to the bracket goes first.** The notation reads right to left, which is exactly why it is worth a mark.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)), & (g \circ f)(x) \\ &= g(f(x)).\end{aligned}$$

These are different functions. With $f(x) = x - 3$ and $g(x) = x^2$:

$$\begin{aligned}(g \circ f)(x) &= (x - 3)^2, & (f \circ g)(x) \\ &= x^2 - 3.\end{aligned}$$

Evaluating at a point. Work inwards out, and carry a *number*, not an expression:

$$\begin{aligned}h(4) &= f(g(4)), \\ g(4) &= 4^2 - 3 \cdot 4 = 4, \\ h(4) &= f(4).\end{aligned}$$

That last line is the whole point of the question: you never need a formula for f , only its value at 4.

A worked example of the same move.

In May 2025 TZ1 Paper 3 a probability distribution is packed into the polynomial

$$G(t) = \frac{1}{16}t + \frac{3}{16}t^2 + \frac{5}{16}t^3 + \frac{7}{16}t^4,$$

and the question asks for $G(1)$. Substituting gives $\frac{1+3+5+7}{16} = 1$ — the probabilities add to one. One mark, one

substitution, and the answer was predictable before any arithmetic.

Composition and the chain rule are the same picture. If a later part asks for $(f \circ g)'(0)$, that is $f'(g(0)) \cdot g'(0)$ — the inner derivative is a factor, not a decoration. B2 stops at the value; the derivative belongs to E3.

Task 1 — evaluating composites

(a) The function f is defined for all $x \in \mathbb{R}$, and the line $y = 6x - 1$ is the tangent to the graph of f at $x = 4$. Given $g(x) = x^2 - 3x$ and $h(x) = f(g(x))$, find $h(4)$.

(b) The function f is defined by $f(x) = \frac{3x - 2}{2x + 1}$, $x \neq -\frac{1}{2}$. Write down the value of $f(0)$.

(c) The function f is defined by $f(x) = e^{2x}(3x - 4)$. A function g satisfies $g(0) = 1$ and $g'(0) = 2$. Find the exact value of $(f \circ g)(0)$.

(d) Two numbers $x_1, x_2 \in \mathbb{R}^+$ satisfy $x_1 + x_2 = 12$. Write their product as a function f of x_1 only. (Enter it in)

Sources: November 2021, Paper 1, Q5(c) (2 marks); May 2025 TZ2, Paper 1, Q1(a) (1 mark); November 2022, Paper 2, Q11(d)(i) (2 marks); May 2023 TZ1, Paper 3, Q2(a) (2 marks).

```
myla = ...                # h(4)
mylb = ...                # f(0)
mylc = ...                # (f ∘ g)(0), exact
myld = ...                # the product as a
function of x

check_num('Task 1(a)', myla, 6,
'535fa30d7e25')
check_num('Task 1(b)', mylb, 6,
'cf3bae39dd69')
verify_exact('Task 1(c)', mylc, -exp(2))
check_series('Task 1(d)', myld,
'6e77545a1577')
```

Theory 2. Building the composite

Substituting a number was arithmetic. Substituting an **expression** is where brackets start earning marks:

$$\begin{aligned}
g(x) &= x^2 + k^2, & f(x) \\
&= x - 3 \\
&\implies (g \circ f)(x) \\
&= (x - 3)^2 + k^2.
\end{aligned}$$

Write the bracket first, expand second. The most common lost mark in this rung is $(x - 3)^2$ turned into $x^2 - 9$, or into $x^2 - 3$.

"Show that" is a different kind of question. The answer is printed in the question, so the marks are for the road to it. A typical three-mark chain:

$$\begin{aligned}
(f \circ g)(x) &= 4^{1+\log_2 x} \\
&= 4 \cdot 4^{\log_2 x} \\
&= 4 \cdot 2^{2\log_2 x} \\
&= 4 \cdot (2^{\log_2 x})^2 \\
&= 4x^2.
\end{aligned}$$

M1 for attempting the composite, A1 for splitting the exponent, A1 for landing on $4x^2$. Writing $4x^2$ straight from the question earns nothing.

Composites hide inside identities too.

$f(t) = \frac{e^t + e^{-t}}{2}$ and $g(t) = \frac{e^t - e^{-t}}{2}$ satisfy $(f(t))^2 + (g(t))^2 = f(2t)$, and the proof is

nothing but substituting the definitions and collecting. Those two functions are $\cosh$ and $\sinh$; the exam does not name them, and does not need to.

Task 2 — write down the composite

Consider $f(x) = x - 3$ and $g(x) = x^2 + k^2$, where k is a real constant. Write down an expression for $(g \circ f)(x)$.

Keep the letter k in your answer — the check substitutes values for it too.

Source: November 2023, Paper 1, Q1(a) (2 marks, no calculator).

```
my2 = ... # (g ∘ f)(x), in x
and k

check_series('Task 2', my2, 'd1d0e04559af')
```

Task 3 — a "show that" with logarithms

The functions are $f(x) = 4^x$, $x \in \mathbb{R}$, and $g(x) = 1 + \log_2 x$, $x \in \mathbb{R}^+$.

Show that $(f \circ g)(x) = 4x^2$.

Enter the composite **before** you simplify it — the unsimplified $4^{1+\log_2 x}$, written in code as `4**(1 + log(x, 2))`. The check compares it with $4x^2$ as functions, which is exactly what a "show that" claims.

Source: May 2025 TZ2, Paper 1, Q10(c) (3 marks, no calculator).

```
my3 = ... # (f ∘ g)(x) as it
comes out, before simplifying

verify_identity('Task 3', my3, 4*x**2,
samples=(0.4, 0.9, 1.6, 2.5, 4.1))
```

Task 4 — a composite identity

The functions f and g are defined for $t \in \mathbb{R}$ by

$$f(t) = \frac{e^t + e^{-t}}{2}, \quad g(t) = \frac{e^t - e^{-t}}{2}.$$

Show that $(f(t))^2 + (g(t))^2 = f(2t)$.

Enter the left-hand side, written out with the definitions substituted and collected as far as you can take it. Use `t`, not `x`.

Source: November 2021, Paper 3, Q1(b)
(3 marks).

```
my4 = ...                                # (f(t))^2 + (g(t))^2,
simplified

verify_identity('Task 4', my4, (exp(2*t) +
exp(-2*t))/2, var=t)
```

Theory 3. The composite is given a value — that is an equation

The moment a composite is set equal to a number, the question stops being about composition and becomes an equation. The technique is the join:

1. build the composite,
2. set it equal to the given value,
3. solve — and then check every root against the domain in the question.

$$\begin{aligned}
(g \circ f)(2) &= 10 \\
&\implies (2 - 3)^2 + k^2 \\
&= 10 \\
&\implies k^2 \\
&= 9 \\
&\implies k \\
&= \pm 3.
\end{aligned}$$

Both signs. "The possible values" is plural on purpose, and $k^2 = 9$ is where the answer is decided.

When f itself is unknown. If $f(x) = ax + b$ and the composite is given as a polynomial, compare coefficients:

$$\begin{aligned}
(g \circ f)(x) &= (ax + b)^2 + (ax + b) + 3 \\
&= 4x^2 - 14x + 15.
\end{aligned}$$

From x^2 : $a^2 = 4$, so $a = \pm 2$. From x : $2ab + a = -14$, which pairs each a with its own b . Two functions come out, not one, and the constant term is the check that both work.

When the domain throws a root away. In a question with $x > 3$ in the definition, a quadratic gives two roots and one of

them is smaller than 3. Rejecting it is a mark; forgetting to look is the single most expensive habit in this rung.

Task 5 — the composite pinned at a point

With $f(x) = x - 3$ and $g(x) = x^2 + k^2$ as in task 2, it is given that $(g \circ f)(2) = 10$. Find the possible values of k .

Enter them as a list.

Source: November 2023, Paper 1, Q1(b) (3 marks, no calculator).

```
my5 = [...]                # the possible
values of k, as a list

verify_root_set('Task 5', my5, k**2 - 9,
var=k)
```

Task 6 — find the function, not the number

The functions f and g are defined for $x \in \mathbb{R}$ by

$$\begin{aligned} f(x) &= ax + b \quad (a, b \in \mathbb{R}), & g(x) \\ &= x^2 + x + 3. \end{aligned}$$

Find the **two** possible functions f such that $(g \circ f)(x) = 4x^2 - 14x + 15$.

Enter both expressions in a list; the order does not matter.

Source: May 2023 TZ2, Paper 1, Q5 (7 marks, no calculator).

```
my6 = [..., ...]          # the two functions
f, as expressions in x

check_set('Task 6', my6, 'e314335912f4')
```

Task 7 — a composite equal to an angle

A function g is defined by $g(x) = \frac{1}{x^2 - 2x - 3}$, where $x \in \mathbb{R}$, $x > 3$. A function h is defined by $h(x) = \arctan \frac{x}{2}$, where $x \in \mathbb{R}$.

Given that $(h \circ g)(a) = \frac{\pi}{4}$, find the value of a . Give your answer in the form $p + \frac{q}{2}\sqrt{r}$, where $p, q, r \in \mathbb{Z}^+$.

The domain $x > 3$ is not decoration here.

*Source: May 2022 TZ2, Paper 1, Q11(c)
(7 marks, no calculator).*

```
my7 = ...                # the value of a,  
exact  
  
verify_exact('Task 7', my7, 1 + 3*sqrt(2)/2)
```

Theory 4. Substituting $-x$: the smallest composite there is

$f(-x)$ is f composed with the function $x \mapsto -x$, and it answers the only symmetry question the exam asks:

$$f(-x) = f(x)$$

$\Rightarrow f$ is even (symmetric in the y -axis),

$$f(-x) = -f(x)$$

$\Rightarrow f$ is odd (symmetric about the origin).

The mark is the sentence. These questions are marked R1 or AG: arriving at $f(-x) = f(x)$ and stopping is not the answer. "Therefore f is even" is.

Write the brackets. $f(-x)$ means replacing every x , brackets included:
 $(-x)^2 = x^2$, $(-x)^3 = -x^3$, $(-x)^4 = x^4$. In

$\arcsin \frac{x^2 - 1}{x^2 + 1}$ every x appears squared, so $f(-x)$ is the same expression, character for character — and that is the proof.

Inside a sum, the claim is about every term. For

$$f_n(x) = \sum_{r=0}^n (-2x^2)^r,$$

each term contains $(-x)^2 = x^2$, so each term is unchanged, so the sum is unchanged. Checking $r = 0$ and $r = 1$ and writing "and so on" is worth less than one line about the general term.

Task 8 — show that it is even

A function f is defined by $f(x) = \arcsin \frac{x^2 - 1}{x^2 + 1}$, $x \in \mathbb{R}$.

Show that f is an even function.

Enter $f(-x)$, written out and simplified; the check compares it with $f(x)$ as functions. Then read the solution for the part the check cannot see — the sentence that earns the mark.

Source: May 2021 TZ2, Paper 2, Q12(a)
(1 mark).

```
my8 = ... # f(-x), simplified  
  
verify_identity('Task 8', my8, asin((x**2 -  
1)/(x**2 + 1)))
```

Task 9 — even for every n

Consider the family of functions

$$f_n(x) = \sum_{r=0}^n (-2x^2)^r, \quad x \in \mathbb{R}, n \in \mathbb{N}.$$

Show that f_n is an even function for all values of n .

The general argument is one line and belongs in your written work. For the check, write out $f_3(-x)$ as a polynomial in ascending powers of x — if the substitution is done correctly it will be $f_3(x)$ itself.

Source: May 2025 TZ1, Paper 2, Q12(a)
(3 marks).

```
my9 = ... # f_3(-x), expanded  
  
check_series('Task 9', my9, '28fe54a8838e')
```

Theory 5. Composing f with itself

f^2 means $f \circ f$, **not** $(f(x))^2$. The notation is unfortunate and the exam knows it — Paper 3 investigations state the convention explicitly before using it:

$$\begin{aligned}f^2(x) &= f(f(x)), \\f^3(x) &= f(f(f(x))), \\f^n(x) &= \underbrace{f(f(\cdots f(x) \cdots))}_{n \text{ times}}.\end{aligned}$$

For a linear $f(x) = mx + c$ the pattern appears immediately, provided you do not simplify it away:

$$f^2(x) = m(mx + c) + c = m^2x + c(1 + m),$$

$$\begin{aligned}f^3(x) &= m(m^2x + c(1 + m)) + c \\&= m^3x + c(1 + m + m^2),\end{aligned}$$

$$f^4(x) = m^4x + c(1 + m + m^2 + m^3).$$

Two separate stories: the coefficient of x is just m^n , and the constant is a geometric sum. Closing it,

$$\begin{aligned}
 f^n(x) &= m^n x + c \\
 (1 + m + \cdots + m^{n-1}) &= m^n x + c \\
 \frac{1 - m^n}{1 - m}, \\
 m &\neq 1.
 \end{aligned}$$

Three cases, and the exam asks about all of them.

- $m = 1$: the closed form is $\frac{0}{0}$ and useless. Go back to the sum: $1 + 1 + \cdots + 1 = n$, so $f^n(x) = x + nc$. The graph slides.
- $-1 < m < 1$: $m^n \rightarrow 0$, so $f^n(x) \rightarrow \frac{c}{1 - m}$ for every x . The family of lines flattens onto the horizontal line $y = \frac{c}{1 - m}$ — which is the fixed point of f .
- $m = -1$: $(-1)^n$ alternates. For odd n , $f^n(x) = -x + c$; for even n , $f^n(x) = x$. The function is **self-inverse**, and $f \circ f$ being the identity is precisely what that means.

That last case is the loop of this practicum closing: composition run twice and coming back to x is the same statement as $f = f^{-1}$.

Task 10 — the self-composite of a linear function

Consider the linear function $f(x) = mx + c$, where $x \in \mathbb{R}$ and $m, c \in \mathbb{R}$.

- (a) Show that $f^2(x) = m^2x + c(1 + m)$.
- (b) Show that $f^3(x) = m^3x + c(1 + m + m^2)$.
- (c) Write down an expression for $f^4(x)$.
- (d) Suggest a similar expression for $f^n(x)$, $n \in \mathbb{Z}^+$, with $m \neq 1$. Write the constant part as a **single fraction**.

Source: May 2025 TZ3, Paper 3, Q1(a)(b)(i)(ii) (3 + 2 + 1 + 2 marks). Part (c) of that question proves the formula by induction — that is A7's work, and this practicum stops at finding it.

```
my10a = ...           #  $f^2(x)$ 
my10b = ...           #  $f^3(x)$ 
my10c = ...           #  $f^4(x)$ 
my10d = ...           #  $f^n(x)$ , constant as
one fraction
```

```
check_series('Task 10(a)', my10a,
'5512df9c2470')
check_series('Task 10(b)', my10b,
'1fd69b5ce511')
check_series('Task 10(c)', my10c,
'f361ce30f981')
check_series('Task 10(d)', my10d,
'664ca5cf1044')
```

Theory 6. The defining property, and how to avoid finding the inverse

Everything about f^{-1} follows from one sentence:

$$f^{-1}(a) = b \iff f(b) = a.$$

Read left to right it is a definition; read right to left it is a technique. Most inverse questions in the archive ask for **one value**, and the value comes out without ever writing the formula:

$$\begin{aligned}
f(x) &= 4^x, & f^{-1}(8) \\
&= ? \\
&\implies 4^x \\
&= 8 \\
&\implies 2^{2x} \\
&= 2^3 \\
&\implies x \\
&= \frac{3}{2}.
\end{aligned}$$

Three corollaries, each worth marks somewhere in the archive.

- **Reflection.** (a, b) on $y = f(x)$ means (b, a) on $y = f^{-1}(x)$. The graphs are mirror images in the line $y = x$, so anything symmetric in that line — a maximum distance, a point of intersection with $y = x$ — is shared by both.
- **Nested.** $f^{-1}(\text{something}) = b$ becomes $\text{something} = f(b)$. In $f^{-1}(2x - 7) = -3$ the inner bracket equals $f(-3)$, **not** -3 . That single confusion is the whole difficulty of the question.
- **Self-inverse.** $g = g^{-1}$ can be attacked two ways: find g^{-1} and equate, or

compute $g(g(x))$ and set it equal to x . Both are in the markscheme; the second is usually shorter.

On a calculator. $f^{-1}(2)$ where $f(x) = 4 \cot x + \sin x$ is the one place in this whole topic where a GDC does real work: solve $f(x) = 2$ on $0 < x < \pi$ and read $x = 1.32$. Everywhere else the "calculator" papers still want exact answers.

Task 11 — one value of the inverse, two ways

(a) The function f is defined by $f(x) = 4^x$, $x \in \mathbb{R}$. Find $f^{-1}(8)$, expressing your answer in the form $\frac{p}{q}$ with $p, q \in \mathbb{Z}$.

(b) The function f is defined by $f(x) = 4 \cot x + \sin x$, where $0 < x < \pi$. Find the value of $f^{-1}(2)$, correct to 3 significant figures.

Part (b) is the calculator question of this practicum. Part (a) is not: $\log_4 8$ is exact and a decimal earns nothing.

Sources: May 2025 TZ2, Paper 1, Q10(a) (3 marks, no calculator); May 2025 TZ1, Paper 2, Q6(b) (1 mark, calculator).

```
my11a = ...                #  $f^{-1}(8)$ , exact  
my11b = ...                #  $f^{-1}(2)$ , 3 s.f.  
  
check_num('Task 11(a)', my11a, 6,  
'9f29a130438b')  
check_num('Task 11(b)', my11b, 3,  
'ec96dbc52385')
```

Task 12 — reading the inverse off a graph

The graph of $y = f(x)$ is drawn for $-6 \leq x \leq 5$. From the diagram in the paper: f is one-to-one on that interval, $f(-3) = -1$, the least value of f is -3 and the greatest is 5 , both attained.

(a) State the domain of f^{-1} .

(b) Find the value of x that satisfies $f^{-1}(2x - 7) = -3$.

Enter the domain as a set or an inequality, for instance `Interval(-3, 5)` or `(x >= -3) & (x <= 5)`.

Source: May 2025 TZ3, Paper 1, Q2(b)(c)
(1 + 3 marks, no calculator).

```
my12a = ...                # domain of  $f^{-1}$ 
my12b = ...                # the value of  $x$ 

check_domain('Task 12(a)', my12a,
'39bc7abc5f32')
check_num('Task 12(b)', my12b, 6,
'4e07408562be')
```

Task 13 — a function equal to its own inverse

The function g is defined by

$$g(x) = \frac{ax + 4}{3 - x}, \quad x \in \mathbb{R}, x \neq 3, a \in \mathbb{R}.$$

Given that $g(x) = g^{-1}(x)$, determine the value of a .

Two routes are in the markscheme. Find g^{-1} and equate it to g , or compute $g(g(x))$ and set it equal to x . Try the second: it turns the question into "which a makes $g \circ g$ the identity", and that is rung 5 in disguise.

Source: November 2021, Paper 1, Q2(d)
(4 marks, no calculator).

```
my13 = ... # the value of a  
check_num('Task 13', my13, 6, '615bdd17c255')
```

Theory 7. Swap, then make y the subject

The formula for an inverse comes from one honest procedure:

1. write $y = f(x)$;
2. **interchange** x and y — this is the mark, and it is the step most often skipped;
3. solve the new equation for y ;
4. call the result $f^{-1}(x)$.

For a rational function, step 3 is always the same three moves — multiply out, collect every y on one side, factorise:

$$\begin{aligned}
x &= \frac{7y + 7}{2y - 4} \\
&\Rightarrow 2xy - 4x \\
&= 7y + 7 \\
&\Rightarrow y(2x - 7) \\
&= 4x + 7 \\
&\Rightarrow f^{-1}(x) \\
&= \frac{4x + 7}{2x - 7}.
\end{aligned}$$

For a logarithm or an exponential, step 3 is one application of the opposite operation:

$$\begin{aligned}
x &= 1 + \log_2 y \\
&\Rightarrow \log_2 y \\
&= x - 1 \\
&\Rightarrow g^{-1}(x) \\
&= 2^{x-1}.
\end{aligned}$$

Check by undoing. Substitute f into your answer and you must get x back. This is not an optional flourish — it is how the checks in this notebook work, and it catches every algebra slip in the collect-and- factorise step:

$$f^{-1}(f(t)) = t \quad \text{for every } t \text{ in the domain of } f.$$

Notation. The answer is $f^{-1}(x) = \dots$, not $x = \dots$ and not $y = \dots$. And f^{-1} is never $\frac{1}{f}$; the notation is borrowed, the meaning is not.

Task 14 — the inverse of a rational function

The function f is defined by $f(x) = \frac{7x + 7}{2x - 4}$ for $x \in \mathbb{R}$, $x \neq 2$.

Find $f^{-1}(x)$, the inverse function of f .

The check substitutes f into whatever you write and requires x back, so an equivalent form is accepted and an algebra slip is not.

Source: May 2023 TZ1, Paper 1, Q1(c) (3 marks, no calculator).

```
my14 = ...                                # f-1(x)

verify_inverse('Task 14', my14, (7*x +
7)/(2*x - 4),
               domain=Interval.open(2, 20))
```

Task 15 — inverse of a logarithm, and what it does to the graph

The function g is defined by $g(x) = 1 + \log_2 x$, where $x \in \mathbb{R}^+$.

(a) Find an expression for $g^{-1}(x)$.

(b) Describe a sequence of transformations that maps the graph of $y = g^{-1}(x)$ onto the graph of $y = f(x)$, where $f(x) = 4^x$.

Part (b) is not checked here — naming transformations in the right order is B3's technique. Write your answer down, then compare it with the solution; the order is the whole mark.

Source: May 2025 TZ2, Paper 1, Q10(b) (4 marks, no calculator).

```
my15 = ...                                #  $g^{-1}(x)$ 

verify_inverse('Task 15', my15, 1 + log(x,
2), domain=Interval.open(0, 12))
```

Theory 8. Where an inverse exists, and on which domain

Existence. f has an inverse exactly when it is one-to-one: no horizontal line meets the graph twice. A function that turns — a parabola, a cosine — is one-to-one only up to its turning point, so the question becomes "where does f turn?"

$$f(x) = \cos(x - k), \quad 0 \leq x \leq a.$$

The graph is a cosine wave shifted right by k . Starting from $x = 0$ it is monotonic until the first turning point, and that point is where $f'(x) = -\sin(x - k)$ first vanishes for $x > 0$. For $\pi < k < 2\pi$ that happens at $x = k - \pi$, so $a = k - \pi$ is the largest domain that keeps an inverse.

The swap. Once the inverse exists, its domain and range are the range and domain of the original — swapped, both of them:

$$\begin{aligned} \text{domain}(f^{-1}) &= \text{range}(f), & \text{range}(f^{-1}) \\ &= \text{domain}(f). \end{aligned}$$

Copying the domain of f across is the standard mistake, and it is usually visible: an inverse whose domain is the same as f 's is suspicious unless the function is self-inverse.

So you have to be able to read a range. In May 2025 TZ2 Paper 1, $f(x) = \frac{3x-2}{2x+1}$ and $g(x) = -f(x)$ on $x \geq 0$. Reflecting in the x -axis sends the horizontal asymptote $y = \frac{3}{2}$ to $y = -\frac{3}{2}$ and the value $f(0) = -2$ to $g(0) = 2$; g decreases from 2 towards $-\frac{3}{2}$ without reaching it, so the range is $-\frac{3}{2} < y \leq 2$. Reading ranges properly is B4's subject; here it is a prerequisite, because the range of f is the answer to "state the domain of f^{-1} ".

Endpoints are marks. $[0, \sqrt{3}]$ and $(0, \sqrt{3})$ are different answers. Whether an endpoint is attained follows from whether the original attains it — f on the closed $[1, 2]$ has a closed range.

Task 16 — the largest domain that keeps an inverse

Let $f(x) = \cos(x - k)$, where $0 \leq x \leq a$ and $a, k \in \mathbb{R}^+$.

(a) For $k = \frac{\pi}{2}$, find the largest value of a for which f^{-1} exists.

(b) Find the largest such a when $k = \pi$.

(c) Find the largest such a when $\pi < k < 2\pi$, giving your answer in terms of k .

Sketch each case before answering — one picture settles all three. Answers are exact.

Source: November 2022, Paper 1, Q8 (2 + 1 + 2 marks, no calculator).

```
myl6a = ...                # largest a for k =
pi/2
myl6b = ...                # largest a for k =
pi
myl6c = ...                # largest a for pi <
k < 2pi, in terms of k

verify_exact('Task 16(a)', myl6a, pi/2)
verify_exact('Task 16(b)', myl6b, pi)
check_expr('Task 16(c)', myl6c,
'08749ab87cf1')
```

Task 17 ● — inverse, domain and range together

Consider the function $f(x) = \sqrt{x^2 - 1}$, where $1 \leq x \leq 2$.

(a) Show that the inverse function of f is $f^{-1}(x) = \sqrt{x^2 + 1}$.

(b) State the domain of f^{-1} .

(c) State the range of f^{-1} .

Enter the inverse in (a) — the check will substitute f into it. For (b) and (c) enter sets: `Interval(0, sqrt(3))`, `Interval(1, 2)`.

Source: May 2022 TZ1, Paper 2, Q10(b) (5 marks). The corpus records a different formula for this answer; the paper is what is used here, and the reason is in the solution.

```

myl7a = ...                #  $f^{-1}(x)$ 
myl7b = ...                # domain of  $f^{-1}$ 
myl7c = ...                # range of  $f^{-1}$ 

verify_inverse('Task 17(a)', myl7a, sqrt(x**2
- 1), domain=Interval(1, 2))
check_domain('Task 17(b)', myl7b,
'b86f8c39da95')
check_domain('Task 17(c)', myl7c,
'4c2477c00b25')

```

Task 18 — justify existence, then find it

The function g is defined by

$$g(x) = \frac{1}{1 + 2x^2},$$

$$0 \leq x < K,$$

where $K = \frac{1}{\sqrt{2}}.$

(a) Justify that g^{-1} exists.

(b) Find $g^{-1}(x)$, giving its domain.

Part (a) is an R1 — one sentence, no algebra — and there is nothing to type; write it down and compare with the solution. For (b), enter the expression and the domain.

Source: May 2025 TZ1, Paper 2, Q12(d) (6 marks). g is the limit of the family f_n from task 9, and K is where that series stops converging.

```
myl8a = ...                #  $g^{-1}(x)$ 
myl8b = ...                # domain of  $g^{-1}$ 

verify_inverse('Task 18(a)', myl8a, 1/(1 +
2*x**2),
              domain=Interval.Ropen(0,
1/sqrt(2)))
check_domain('Task 18(b)', myl8b,
'8b22960c3254')
```

Theory 9. Choosing the branch — the R1 at the end of the root

Whenever the swap ends in $y^2 = \text{something}$, the algebra offers two functions and only one of them is the inverse:

$$y = \pm \sqrt{\text{something}}.$$

What chooses. The values of f^{-1} are the inputs of f , so

$$\text{range}(f^{-1}) = \text{domain}(f),$$

and you keep the branch whose values lie in the domain of f . If f was defined on $x > 3$, then f^{-1} must take values above 3, and the negative root cannot. If f was defined on $x \geq 0$, the positive root survives.

Note which set decides: the **range** of f^{-1} , not its domain. They are different sets, and only one of them is about the sign.

Say it. The mark is R1, awarded for the sentence "the negative root is rejected because the domain of g is $x > 3$ ". A correct final formula with no reason loses it.

Why checking by $f(f^{-1}(x)) = x$ will not save you. Take $f(x) = \sqrt{x^2 - 1}$ on $[1, 2]$ and the wrong branch $g(x) = -\sqrt{x^2 + 1}$. Then

$$f(g(x)) = \sqrt{\left(-\sqrt{x^2 + 1}\right)^2 - 1} = \sqrt{x^2} = x$$

for $x \geq 0$ — the square erases the sign, and the wrong branch passes. Compose the other way and it fails at once:

$$g(f(t)) = -\sqrt{t^2 - 1 + 1} = -t \neq t.$$

That is why every inverse in this notebook is checked as $f^{-1}(f(t)) = t$, and it is the same asymmetry the R1 is testing.

Task 19 ● — completing the square inside an inverse

A function g is defined by

$$g(x) = \frac{1}{x^2 - 2x - 3}, \quad x \in \mathbb{R}, x > 3.$$

(a) Show that $g^{-1}(x) = 1 + \frac{\sqrt{4x^2 + x}}{x}$.

(b) State the domain of g^{-1} .

Completing the square is the move: $y^2 - 2y - 3 = (y - 1)^2 - 4$. Then say out loud which branch you keep and why — and remember that $\sqrt{\frac{4x+1}{x}} = \frac{\sqrt{4x^2+x}}{x}$ only because $x > 0$.

Source: May 2022 TZ2, Paper 1, Q11(b) (7 marks, no calculator).

```

my19a = ...                #  $g^{-1}(x)$ 
my19b = ...                # domain of  $g^{-1}$ 

verify_inverse('Task 19(a)', my19a, 1/(x**2 -
2*x - 3),
              domain=Interval.open(3, 20))
check_domain('Task 19(b)', my19b,
'0ed23e410b98')

```

Task 20 — an inverse through arcsin

A function g is defined by

$$g(x) = \arcsin \frac{x^2 - 1}{x^2 + 1}, \quad x \in \mathbb{R}, x \geq 0$$

— the function of task 8, restricted to $x \geq 0$ so that an inverse exists.

(a) Find an expression for $g^{-1}(x)$, justifying your answer.

(b) State the domain of g^{-1} .

Apply $\sin$ to both sides after the swap, then solve for y^2 . The justification in (a) is the R1 of this rung.

Source: May 2021 TZ2, Paper 2, Q12(d) (e) (5 + 1 marks).

```

my20a = ...                #  $g^{-1}(x)$ 
my20b = ...                # domain of  $g^{-1}$ 

verify_inverse('Task 20(a)', my20a,
asin((x**2 - 1)/(x**2 + 1)),
          domain=Interval(0, oo))
check_domain('Task 20(b)', my20b,
'28c48aa1ddb7')

```

Trainer: name the technique in five seconds

Fifteen questions with no solutions. For each one write the code of the technique — nothing else. No calculation is wanted; the point is to find the first move before the arithmetic starts.

Codes: `comp` (evaluate a composite) · `build` (build the composite) · `ceq` (composite set equal to something) · `even` (substitute $-x$) · `iter` (compose f with itself) · `prop` (the defining property of the inverse) · `swap` (swap and solve) · `dom` (existence and domain) · `branch` (choose the branch)

1. Given $f(x) = 2x + 1$ and $g(x) = x^2$, write down $(g \circ f)(x)$.
2. $f(x) = 3x - 5$. Find $f^{-1}(x)$.
3. Given $f(x) = x^3 + x$, find $f^{-1}(10)$.
4. $h(x) = x^2 - 4x$ and $k(x) = \sqrt{x}$. Find $(k \circ h)(6)$.
5. Show that $f(x) = x^4 - 3x^2 + 1$ is an even function.
6. $f(x) = \ln(x - 2)$, $x > 2$, has range $\mathbb{R}$. State the domain of f^{-1} .
7. $f(x) = mx + c$. Find $f(f(x))$.
8. $(f \circ g)(x) = x^2 + 6x + 7$ and $g(x) = x + 3$. Find $f(x)$.
9. $g(y) = y^2 - 6y$ for $y \leq 3$. Find g^{-1} , justifying your choice of sign.
10. $f(x) = x^2 - 6x + 5$, $0 \leq x \leq a$. Find the largest a for which f^{-1} exists.
1. $f(x) = \frac{x+2}{x-1}$, $x \neq 1$. Find $f^{-1}(x)$.
2. $f(x) = \frac{1}{x}$, $x \neq 0$. Find $f^3(x)$.
3. $f(x) = x - 1$ and $g(x) = x^2 + a$, and $(g \circ f)(3) = 7$. Find a .

4. Show that $f(x) = \frac{x}{x^2 + 1}$ is an odd function.

5. The graph of f passes through $(2, 7)$. Write down a point on the graph of f^{-1} .

```
answers = {
    1: '',    2: '',    3: '',    4: '',    5:
    '',
    6: '',    7: '',    8: '',    9: '',    10:
    '',
    11: '',   12: '',   13: '',   14: '',   15:
    '',
}

trigger_check(answers, {1: '44575cf5b285', 2:
'da47c2f450a4', 3: 'e835d0bf51a4', 4:
'f562267f2b6d', 5: 'ec38063c881f', 6:
'7e6e9877fff0', 7: '82439cbebbb0', 8:
'2eb8951883c1', 9: 'f38c764c8aa0', 10:
'7e6e9877fff0', 11: 'da47c2f450a4', 12:
'82439cbebbb0', 13: '2eb8951883c1', 14:
'ec38063c881f', 15: 'e835d0bf51a4'})
```

Task 21 — on the timer, 17 minutes

Part of a Paper 3 investigation. Eleven marks, target time about 17 minutes. You have already met the setup in task 10, so

nothing here is repeated: this is what the investigation does with the formula once it has it.

For $f(x) = mx + c$ it is given that

$$f^n(x) = m^n x + c \frac{1 - m^n}{1 - m}, \quad m \neq 1.$$

(b)(iii) By using your expression from part (b)(ii), or otherwise, find an expression in terms of n for $f^n(x)$ when $m = 1$. [3]

(d) Now let $-1 < m < 1$. As $n \rightarrow \infty$, the family of graphs $y = f^n(x)$ approaches the graph of a straight line L . Determine the equation of L , giving your answer in terms of c and m . Enter the right-hand side only. [4]

(e)(i) Now let $m = -1$. Show that $f^n(x) = -x + c$ when n is odd. Enter $f^n(x)$ for odd n . [2]

(e)(ii) Find an expression for $f^n(x)$ when n is even. [2]

Source: May 2025 TZ3, Paper 3, Q1(b)(iii)(d)(e) (3 + 4 + 2 + 2 marks). Part (d)

is a limit, which belongs to E-series work; it is here because the investigation asks for it in the same breath.

```
my21a = ...                #  $f^n(x)$  when  $m = 1$ 
my21b = ...                # the right-hand
side of the equation of L
my21c = ...                #  $f^n(x)$  when  $m = -1$ 
and  $n$  is odd
my21d = ...                #  $f^n(x)$  when  $m = -1$ 
and  $n$  is even

check_series('Task 21(b)(iii)', my21a,
'd1172d76ed9b')
check_expr('Task 21(d)', my21b,
'cd8024d7e992')
check_series('Task 21(e)(i)', my21c,
'116abce4acc4')
check_series('Task 21(e)(ii)', my21d,
'88776d8b0c1a')
```


Solutions

A full discussion of every task, with the markscheme breakdown. Open it after you have worked the task yourself: the point is not the answer but where the marks are, and which of them a self-check cannot see.

Solution 1 — evaluating composites

(a) The tangent to f at $x = 4$ is $y = 6x - 1$, so the point of tangency is on both graphs:

$$f(4) = 6 \cdot 4 - 1 = 23 \quad (\text{and } f'(4) = 6).$$

Then $g(4) = 4^2 - 3 \cdot 4 = 4$, so

$$h(4) = f(g(4)) = f(4) = 23.$$

$$\textbf{(b)} \quad f(0) = \frac{3 \cdot 0 - 2}{2 \cdot 0 + 1} = -2.$$

$$\textbf{(c)} \quad (f \circ g)(0) = f(g(0)) = f(1) = e^2(3 - 4) = -e^2 \approx -7.39.$$

(d) From $x_1 + x_2 = 12$ we get $x_2 = 12 - x_1$, so the product is

$$f(x_1) = x_1(12 - x_1).$$

```
myla = 23
mylb = -2
mylc = -exp(2)
myld = x*(12 - x)
```

Marks. (a) M1 for writing $h(4) = f(g(4))$, A1 for 23. (b) A1 (the pair $(0, -2)$ is accepted). (c) M1 for composing in the correct order, A1 for $-e^2$. (d) M1 for $x_2 = 12 - x_1$, A1 for the product.

The joke in (a). $g(4) = 4$: the inner function sends 4 back to itself. So the only thing you ever need to know about f is $f(4)$, and the tangent line hands it over. Candidates who set out to find a formula for f lost the question to the clock.

What comes next in (a). Part (d) of that question asks for the tangent to h at $x = 4$. That needs $h'(4) = f'(g(4))g'(4) = 6 \cdot 5 = 30$, giving $y = 30(x - 4) + 23$. Composition and the chain rule are the same picture seen twice.

What comes next in (c). The following part asks for $(f \circ g)'(0) = f'(g(0))g'(0) =$

$f'(1) \cdot 2$. With $f'(x) = e^{2x}(6x - 5)$ that is $2e^2 \approx 14.8$ — and the factor $g'(0) = 2$ is precisely the mark.

Solution 2 — write down the composite

$(g \circ f)$ means f first:

$$\begin{aligned}(g \circ f)(x) &= g(x - 3) \\ &= (x - 3)^2 + k^2 \\ &= x^2 - 6x + 9 + k^2.\end{aligned}$$

```
my2 = (x - 3)**2 + k**2
```

Marks. M1 for attempting the composite, A1 for the expression. Both the bracketed and the expanded form are accepted.

The other order is a different function. $(f \circ g)(x) = g(x) - 3 = x^2 + k^2 - 3$. It is just as plausible on the page and worth nothing.

Brackets. $(x - 3)^2$ is $x^2 - 6x + 9$. The two ways to lose the A1 here are $x^2 - 9$ and $x^2 - 3$, and both come from substituting without writing the bracket first.

Solution 3 — a "show that" with logarithms

$$\begin{aligned}(f \circ g)(x) &= f(1 + \log_2 x) \\ &= 4^{1 + \log_2 x} \\ &= 4 \cdot 4^{\log_2 x}.\end{aligned}$$

Now rewrite base 4 as base 2:

$$4^{\log_2 x} = (2^2)^{\log_2 x} = 2^{2 \log_2 x} = (2^{\log_2 x})^2 = x^2,$$

so $(f \circ g)(x) = 4x^2$, as required.

```
my3 = 4**(1 + log(x, 2))
```

Marks. M1 for attempting the composite, A1 for splitting the exponent as $4 \cdot 4^{\log_2 x}$, A1 for landing on $4x^2$.

The natural-log route. $4^{\log_2 x} = e^{\ln 4 \cdot \log_2 x} = e^{2 \ln 2 \cdot \frac{\ln x}{\ln 2}} = e^{2 \ln x} = x^2$. Same three lines, no cleverness needed.

Order, again. $(g \circ f)(x) = 1 + \log_2(4^x) = 1 + 2x$ — a straight line, not a parabola. The two composites of these two functions could hardly look less alike, and that is why the M1 is for the order.

Why enter the unsimplified form? Because a "show that" is a claim about a

transition, not about an answer. The check compares your starting expression with the printed result as *functions*; if they agree everywhere, your first line was right, and the rest is exponent arithmetic you can see on the page.

Solution 4 — a composite identity

Substituting the definitions and putting everything over 4:

$$(f(t))^2 + (g(t))^2 = \frac{(e^t + e^{-t})^2 + (e^t - e^{-t})^2}{4}.$$

Expanding, the cross terms cancel:

$$\begin{aligned} &= \frac{(e^{2t} + 2 + e^{-2t}) + (e^{2t} - 2 + e^{-2t})}{4} = \frac{2e^{2t} + 2e^{-2t}}{4} \\ &= \frac{e^{2t} + e^{-2t}}{2} \\ &= f(2t). \end{aligned}$$

```
my4 = (exp(2*t) + exp(-2*t))/2
```

Marks. METHOD 1: M1 for substituting f and g , M1 for the single fraction, A1 for the simplification — then AG. METHOD 2 runs backwards from $f(2t) = \frac{(e^t)^2 + (e^{-t})^2}{2}$ and

is worth the same; mixtures of the two are accepted.

What these functions are. $f = \cosh$ and $g = \sinh$, and the identity is $\cosh^2 t + \sinh^2 t = \cosh 2t$. The exam does not name them and does not have to — everything follows from the two exponentials.

Careful which identity. The famous one is $\cosh^2 - \sinh^2 = 1$, with a minus. Here the sign is plus and the answer is $f(2t)$, not 1. Reaching for the remembered identity instead of expanding is the way to lose all three marks.

Solution 5 — the composite pinned at a point

Substitute $x = 2$ into the composite from task 2:

$$\begin{aligned}
 (2 - 3)^2 + k^2 &= 10 \\
 \implies 1 + k^2 &= 10 \\
 \implies k^2 &= 9 \\
 \implies k &= \pm 3.
 \end{aligned}$$

my5 = [-3, 3]

Marks. M1 for substituting $x = 2$ into *their* composite and setting it equal to 10, (A1) for $k^2 = 9$, A1 for $k = \pm 3$.

Both signs. k is a real constant with no sign restriction, and the question says "values", plural. $k^2 = 9$ is where the answer is decided, and stopping at $k = 3$ costs the final A1.

The M1 follows your own work. If task 2 came out wrong, this mark is still available — the examiner substitutes into whatever composite you wrote. Which is another reason to write the composite down as a separate line rather than in your head.

Solution 6 — find the function, not the number

With $f(x) = ax + b$,

$$\begin{aligned}(g \circ f)(x) &= (ax + b)^2 + (ax + b) + 3 \\ &= a^2x^2 \\ &\quad + (2ab + a)x \\ &\quad + (b^2 + b + 3) .\end{aligned}$$

Comparing with $4x^2 - 14x + 15$:

$$a^2 = 4 \Rightarrow a = \pm 2, \quad 2ab + a = -14.$$

Each a carries its own b :

$$\begin{aligned}a = 2 : 4b + 2 \\ &= -14 \\ &\Rightarrow b \\ &= -4, \quad a \\ &= -2 : -4b - 2 \\ &= -14 \\ &\Rightarrow b \\ &= 3.\end{aligned}$$

The constant term checks both: $(-4)^2 - 4 + 3 = 15$ and $3^2 + 3 + 3 = 15$.

$$f(x) = 2x - 4 \quad \text{or} \quad f(x) = -2x + 3.$$

$$\text{my6} = [2*x - 4, -2*x + 3]$$

Marks. M1 for attempting $(g \circ f)(x)$, A1 for the expanded form, M1 for equating corresponding terms, A1 for $a = \pm 2$, M1 for using $2ab + a = -14$ to pair the values, A1A1 for the two functions.

Pairing is the whole difficulty. $a = \pm 2$ and $b \in \{-4, 3\}$ is not the answer: it suggests four functions, and only two of them work. The equation $2ab + a = -14$ is what ties each a to its own b , and the mark scheme gives that step its own M1.

The constant term is free verification. It was never used to find anything, so if it comes out as 15 for both pairs, the work is right. Two lines of arithmetic, and the question is closed.

Solution 7 — a composite equal to an angle

$$\begin{aligned}(h \circ g)(a) &= h(g(a)) \\&= \arctan \frac{g(a)}{2} \\&= \frac{\pi}{4} \\&\implies \frac{g(a)}{2} \\&= \tan \frac{\pi}{4} \\&= 1 \\&\implies g(a) \\&= 2.\end{aligned}$$

Now unwind g :

$$\begin{aligned}\frac{1}{a^2 - 2a - 3} &= 2 \\&\Rightarrow a^2 - 2a - 3 \\&= \frac{1}{2} \\&\Rightarrow 2a^2 - 4a - 7 \\&= 0,\end{aligned}$$

$$\begin{aligned}
 a &= \frac{4 \pm \sqrt{16 + 56}}{4} \\
 &= \frac{4 \pm \sqrt{72}}{4} \\
 &= \frac{4 \pm 6\sqrt{2}}{4} \\
 &= 1 \pm \frac{3}{2}\sqrt{2}.
 \end{aligned}$$

The domain of g is $x > 3$, and $1 - \frac{3}{2}\sqrt{2} \approx -1.12$ is not in it. So

$$a = 1 + \frac{3}{2}\sqrt{2} \approx 3.12.$$

```
my7 = 1 + 3*sqrt(2)/2
```

Marks. M1 for attempting $(h \circ g)(a)$, A1 for $\arctan \frac{g(a)}{2}$, M1 for solving for $g(a)$, A1 for $g(a) = 2$, then A1 for the quadratic, M1 for solving it and A1 for the surd form.

Two unwrappings, one after the other. First $\arctan$ (apply $\tan$), then g (multiply up). Each is a step the markscheme pays for separately, and neither needs the formula for g^{-1} — although $a = g^{-1}(2)$ is an accepted route, and task 19 finds that formula.

$\sqrt{72}$ **has to be simplified.** The answer is required in the form $p + \frac{q}{2}\sqrt{r}$ with positive integers, so $\frac{4+\sqrt{72}}{4}$ is not yet an answer: $\sqrt{72} = 6\sqrt{2}$ gives $1 + \frac{3}{2}\sqrt{2}$, that is $p = 1$, $q = 3$, $r = 2$.

The rejected root is a mark. Not because it is ugly, but because g was only ever defined for $x > 3$. Writing "reject $1 - \frac{3}{2}\sqrt{2}$ since $a > 3$ " is the sentence the examiner is looking for.

Solution 8 — show that it is even

$$\begin{aligned} f(-x) &= \arcsin \frac{(-x)^2 - 1}{(-x)^2 + 1} \\ &= \arcsin \frac{x^2 - 1}{x^2 + 1} \\ &= f(x), \end{aligned}$$

therefore f is an even function.

```
my8 = asin((x**2 - 1)/(x**2 + 1))
```

Marks. R1, and it is for the conclusion. A sketch showing symmetry in the y -axis, with the symmetry indicated, is also accepted.

Every x appears squared. That is the entire content: $(-x)^2 = x^2$, so nothing in the expression changes. When a function is built only from even powers of x , its parity is visible before any algebra.

What the check cannot see. It compares $f(-x)$ with $f(x)$ as expressions and says they agree. The mark is for writing the sentence "therefore f is even" on your paper. No self-check in this notebook can award an R1; that is what the solutions are for.

A note on the archive. The corpus records this function upside down, as $\arcsin \frac{1-x^2}{1+x^2}$. Parity survives the misprint — both versions are even — but the inverse does not, which is why task 20 uses the function as the paper prints it.

Solution 9 — even for every n

Write the sum out for $n = 3$:

$$f_3(x) = \sum_{r=0}^3 (-2x^2)^r = 1 - 2x^2 + 4x^4 - 8x^6.$$

Now the general argument, which is what the three marks are for. Replacing x by $-x$ inside the general term,

$$(-2(-x)^2)^r = (-2x^2)^r,$$

so **every** term of the sum is unchanged, and therefore

$$\begin{aligned} f_n(-x) &= \sum_{r=0}^n (-2(-x)^2)^r \\ &= \sum_{r=0}^n (-2x^2)^r \\ &= f_n(x) \end{aligned}$$

for every n . Hence f_n is even.

```
my9 = 1 - 2*x**2 + 4*x**4 - 8*x**6
```

Marks. M1 for attempting to replace x by $-x$, A1 for a correct expression for $f_n(-x)$, A1 for reaching $f_n(x)$ — then AG.

"For all values of n " is the point. Checking $n = 3$ is an example, not a proof. The one line about the general term does every n at once, and it is shorter than writing out a single case.

Where this family goes. f_n is a geometric series with ratio $-2x^2$; it converges exactly when $|2x^2| < 1$, that is $|x| < \frac{1}{\sqrt{2}}$, and its sum is $\frac{1}{1+2x^2}$. That function, on $0 \leq x < \frac{1}{\sqrt{2}}$, is the g of task 18 — and $K = \frac{1}{\sqrt{2}}$ is where the series stops converging.

Solution 10 — the self-composite of a linear function

(a) $f^2(x) = f(f(x)) = m(mx + c) + c = m^2x + mc + c = m^2x + c(1 + m).$

(b) $f^3(x) = f(f^2(x)) = m(m^2x + c(1 + m)) + c = m^3x + cm + cm^2 + c = m^3x + c(1 + m + m^2).$

(c) The pattern is now visible: $f^4(x) = m^4x + c(1 + m + m^2 + m^3).$

(d) In general

$$\begin{aligned}
 f^n(x) &= m^n x \\
 &+ c(1 + m + \cdots + m^{n-1}) = m^n x \\
 &+ c \\
 &\frac{1 - m^n}{1 - m}, \\
 &m \neq 1.
 \end{aligned}$$

```

myl0a = m**2*x + c*(1 + m)
myl0b = m**3*x + c*(1 + m + m**2)
myl0c = m**4*x + c*(1 + m + m**2 + m**3)
myl0d = m**n*x + c*(1 - m**n)/(1 - m)

```

Marks. (a) M1 for attempting f^2 , A1 for $m(mx + c) + c$, A1 for $m^2x + cm + c$, then AG. (b) M1 and A1, then AG. (c) A1, equivalent forms accepted. (d) A1 for $m^n x$ and A1 for the constant — the two halves are marked separately, because they are two different observations.

Do not tidy too early. $f^2(x) = m^2x + mc + c$ is where the pattern lives; collapsing it to a single number for particular m and c destroys the question. The instruction "show that $f^2(x) = m^2x + c(1 + m)$ " is telling you which grouping to keep.

Two sequences, not one. The coefficient of x is a plain geometric sequence m^n . The constant is a geometric sum, and closing it with $\frac{1-m^n}{1-m}$ is A2's formula doing the work. Recognising that the constant is a sum is what makes part (d) two marks rather than one.

The exclusion $m \neq 1$ is real. At $m = 1$ the closed form reads $\frac{0}{0}$. The sum itself is fine — it is $1 + 1 + \cdots + 1 = n$ — which is exactly what the timed task asks about.

And the proof? Part (c) of the original question proves the formula by induction, for eight marks. That is A7's technique; this practicum stops at finding the pattern, which is the harder half to teach and the easier half to do.

Solution 11 — one value of the inverse, two ways

(a) $f^{-1}(8)$ is the x with $f(x) = 8$:

$$\begin{aligned} 4^x &= 8 \\ \iff 2^{2x} &= 2^3 \\ \iff 2x &= 3 \\ \iff x &= \frac{3}{2}. \end{aligned}$$

Equivalently $f^{-1}(8) = \log_4 8 = \frac{\ln 8}{\ln 4} = \frac{3 \ln 2}{2 \ln 2} = \frac{3}{2}$.

(b) $f^{-1}(2)$ is the solution of $4 \cot x + \sin x = 2$ on $0 < x < \pi$. On a GDC, $x = 1.31837 \dots \approx 1.32$.

```
mylla = Rational(3, 2)
myllb = 1.32
```

Marks. (a) A1 for turning the question into $4^x = 8$ (or into $\log_4 8$), M1 for using a common base or a change of base, A1 for $\frac{3}{2}$. (b) A1 for 1.32.

Neither part needs the inverse function. $f^{-1}(a) = b \iff f(b) = a$ turns both into ordinary equations. Deriving $f^{-1}(x) = \log_4 x$ first is not wrong, just longer; deriving an inverse for $4 \cot x + \sin x$ is not possible at all.

Why does f^{-1} even exist in (b)?
Because

$$f'(x) = \cos x - \frac{4}{\sin^2 x} < 0 \quad \text{for } 0 < x < \pi,$$

since $\frac{4}{\sin^2 x} \geq 4$ while $\cos x \leq 1$. So f is strictly decreasing on the whole interval, hence one-to-one. The question does not ask, but this is the reasoning rung 8 makes explicit.

The one honest calculator mark in this practicum. Everything else in the "calculator" papers of this topic wants an exact expression. Here the answer genuinely is a decimal, and it wants three significant figures.

Solution 12 — reading the inverse off a graph

(a) The domain of f^{-1} is the range of f . From the graph f takes every value from -3 to 5 and attains both, so

$$\text{domain}(f^{-1}) = -3 \leq x \leq 5.$$

(b) Apply f to both sides of $f^{-1}(2x - 7) = -3$:

$$\begin{aligned} 2x - 7 &= f(-3) \\ &= -1 \\ \implies 2x &= 6 \\ \implies x &= 3. \end{aligned}$$

```
my12a = Interval(-3, 5)
my12b = 3
```

Marks. (a) A1; interval notation $[-3, 5]$ is accepted. (b) M1 for $2x - 7 = f(-3)$ (or for $f^{-1}(-1) = -3$), A1 for $2x - 7 = -1$, A1 for $x = 3$.

The trap in (b). Setting $2x - 7 = -3$ gives $x = 2$ and no marks. The -3 on the right of $f^{-1}(\dots) = -3$ is an *output* of f^{-1} , so it is an *input* of f ; the thing equal to $2x - 7$ is $f(-3)$. Writing the property out in full before substituting anything is the way to keep this straight.

Why (a) is one mark and not three. Because on this rung the work is reading a range, not computing one. The graph does it for you. In a question without a picture — task 17, task 18 — finding the range is most of the work, and the marks move accordingly.

Solution 13 — a function equal to its own inverse

METHOD 2 (shorter). $g = g^{-1}$ says exactly that $g(g(x)) = x$. Compute:

$$\begin{aligned}
 g(g(x)) &= \frac{a \cdot \frac{ax+4}{3-x} + 4}{3 - \frac{ax+4}{3-x}} \\
 &= \frac{a(ax+4) + 4(3-x)}{3(3-x) - (ax+4)} \\
 &= \frac{(a^2-4)x + 4a + 12}{-(a+3)x + 5}.
 \end{aligned}$$

Setting that equal to x and clearing the denominator,

$$(a^2 - 4)x + 4a + 12 = -(a + 3)x^2 + 5x.$$

The coefficient of x^2 must vanish: $a + 3 = 0$, so $a = -3$. The other two coefficients agree with that value ($a^2 - 4 = 5$ and $4a + 12 = 0$), which confirms it.

METHOD 1. Swap and solve: $y = \frac{ax+4}{3-x}$ gives $3y - xy = ax + 4$, so $x(a + y) = 3y - 4$ and

$$g^{-1}(x) = \frac{3x - 4}{x + a}.$$

Equating $\frac{ax+4}{3-x} \equiv \frac{3x-4}{x+a}$ and comparing coefficients gives $a = -3$ again.

my13 = -3

Marks. METHOD 1: M1 for attempting x in terms of y , A1 for the rearrangement,

A1 for g^{-1} , A1 for $a = -3$. METHOD 2: M1 for attempting $g(g(x))$ and equating to x , A1 for the composite, A1 for the cleared equation, A1 for equating coefficients of x^2 .

Check it. With $a = -3$, $g(x) = \frac{4-3x}{3-x}$. Then $g(0) = \frac{4}{3}$ and $g(\frac{4}{3}) = \frac{4-4}{3-\frac{4}{3}} = 0$. The pair $(0, \frac{4}{3})$ and $(\frac{4}{3}, 0)$ is the reflection in $y = x$ that self-inverse means.

A fact worth carrying. For any $g(x) = \frac{\alpha x + \beta}{\gamma x + \delta}$, the composite $g(g(x))$ is the identity exactly when $\alpha + \delta = 0$ (or g is already the identity). Here $\alpha = a$ and $\delta = 3$, so $a = -3$ in one line. The exam will not give marks for quoting it, but it tells you in advance what the answer must be.

And a warning about the archive. The corpus records this function with its denominator reversed, as $\frac{\alpha x + 4}{x - 3}$. That is a different question, and it has a different answer: $a = +3$. This is the one place in the topic where a mangled formula also carries a mangled answer, and the only

defence is to derive the value rather than remember it.

Solution 14 — the inverse of a rational function

Write $y = f(x)$, interchange x and y , and solve:

$$\begin{aligned}x &= \frac{7y + 7}{2y - 4} \\&\implies x(2y - 4) \\&= 7y + 7 \\&\implies 2xy - 4x \\&= 7y + 7,\end{aligned}$$

$$2xy - 7y = 4x + 7 \implies y(2x - 7) = 4x + 7,$$

$$f^{-1}(x) = \frac{4x + 7}{2x - 7}, \quad x \neq \frac{7}{2}.$$

```
my14 = (4*x + 7)/(2*x - 7)
```

Marks. M1 for interchanging x and y (seen anywhere), A1 for correct working with the y terms on the same side, A1 for the final expression.

The three moves never change. Multiply out, collect every y on one side,

factorise. The factorisation is the step people skip, and without it there is a y left on the right and the "answer" is not a function.

Where does $x \neq \frac{7}{2}$ come from? It is the value that kills the new denominator, and it is f 's horizontal asymptote: $f(x) \rightarrow \frac{7}{2}$ as $x \rightarrow \pm\infty$, so $\frac{7}{2}$ is the one value f never takes, hence the one value f^{-1} cannot be given. The excluded point of an inverse is always the asymptote of the original.

Check by undoing. $f^{-1}(f(t))$: substitute and everything cancels back to t . That is what the check in the cell above does, symbolically here — for a rational function `simplify` finishes the job on its own.

Solution 15 — inverse of a logarithm, and what it does to the graph

(a)
$$x = 1 + \log_2 y \Rightarrow \log_2 y = x - 1 \Rightarrow g^{-1}(x) = 2^{x-1}.$$

(b) Write the target in base 2: $f(x) = 4^x = 2^{2x}$. Starting from $y = 2^{x-1}$:

$$2^{x-1} \xrightarrow{\text{translate 1 left}} 2^x \xrightarrow{\text{horizontal stretch, factor } \frac{1}{2}} 2^{2x} = 4^x.$$

```
my15 = 2**(x - 1)
```

Marks. M1 for interchanging x and y , A1 for $g^{-1}(x) = 2^{x-1}$, A1 for one correct transformation, A1 for both in the correct order.

The order carries the last mark. Do the stretch first and 2^{x-1} becomes 2^{2x-1} ; translating left by 1 then gives 2^{2x+1} , which is $2 \cdot 4^x$, not 4^x . Two correct transformations in the wrong order are worth one mark out of two.

Why g^{-1} is an exponential. Because g was a logarithm, and the two are inverse operations. The "1+" outside the logarithm becomes the "-1" in the exponent — an outside shift on g turns into an inside shift on g^{-1} , which is the reflection in $y = x$ doing its work.

Naming transformations is B3. The check above only looks at the formula. Getting the sequence and its order right is a technique of its own, and this question is where the two topics touch.

Solution 16 — the largest domain that keeps an inverse

(a) $k = \frac{\pi}{2}$: $f(x) = \cos\left(x - \frac{\pi}{2}\right) = \sin x$, which increases on $\left[0, \frac{\pi}{2}\right]$ and turns at $\frac{\pi}{2}$. So $a = \frac{\pi}{2}$.

(b) $k = \pi$: $f(x) = \cos(x - \pi) = -\cos x$, which increases on $[0, \pi]$ and turns at π . So $a = \pi$.

(c) $\pi < k < 2\pi$: differentiate, $f'(x) = -\sin(x - k)$, and at the left endpoint $f'(0) = -\sin(-k) = \sin k < 0$, so f starts by decreasing. It keeps decreasing until f' vanishes, that is until $\sin(x - k) = 0$; the first such $x > 0$ is $x - k = -\pi$, i.e. $x = k - \pi$, which lies in $(0, \pi)$. So $a = k - \pi$.

```
myl6a = pi/2
myl6b = pi
myl6c = k - pi
```

Marks. (a) A2 (A1A0 for a correct sketch with the wrong a). (b) A1. (c) A1 for a sketch showing the wave decreasing as it crosses the y -axis, A1 for $a = k - \pi$.

One picture answers all three. $\cos(x - k)$ is the cosine wave slid k to the right. Draw it, mark $x = 0$, and read off where the curve first turns to the right of that mark. Every part of this question is that one reading.

"Largest a ", not "largest interval". The domain is forced to start at 0. There are longer intervals on which $\cos(x - k)$ is one-to-one — they just do not begin where the question begins.

Endpoints are safe here. At the turning point the function is still one-to-one on the closed interval: it takes each value once and stops. Including $x = a$ is correct, and the markscheme prints it.

Solution 17 — inverse, domain and range together

(a) $y = \sqrt{x^2 - 1}$; interchange and solve:

$$\begin{aligned}
 x &= \sqrt{y^2 - 1} \\
 &\Rightarrow x^2 \\
 &= y^2 - 1 \\
 &\Rightarrow y^2 \\
 &= x^2 + 1 \\
 &\Rightarrow y \\
 &= \sqrt{x^2 + 1},
 \end{aligned}$$

the positive root because the values of f^{-1} lie in $[1, 2]$, the domain of f .

(b) The domain of f^{-1} is the range of f . On $[1, 2]$ the function $\sqrt{x^2 - 1}$ increases from $f(1) = 0$ to $f(2) = \sqrt{3}$, both attained, so

$$\text{domain}(f^{-1}) = [0, \sqrt{3}].$$

(c) The range of f^{-1} is the domain of f , that is $[1, 2]$.

```

my17a = sqrt(x**2 + 1)
my17b = Interval(0, sqrt(3))
my17c = Interval(1, 2)

```

Marks. M1 for interchanging (seen anywhere), A1 for $x^2 = y^2 - 1$, A1 for $y = \sqrt{x^2 + 1}$, then AG; A1 for the domain, A1 for the range.

The wrong branch passes the wrong test. $-\sqrt{x^2 + 1}$ satisfies $f(g(x)) = \sqrt{(-\sqrt{x^2 + 1})^2 - 1} = x$ for $x \geq 0$: the square erases the sign. Composed the other way it fails immediately, $g(f(t)) = -t$. This is the question that fixed the direction of the check used throughout this notebook.

Closed brackets, both ends. f is defined on a closed interval and is continuous, so it attains both 0 and $\sqrt{3}$. Writing $(0, \sqrt{3})$ costs the mark, and the difference is not a technicality — it is whether $f^{-1}(0)$ is defined.

A note on the archive. The corpus records this answer as $f^{-1}(x) = \frac{2}{x^2} + 1$, and the function itself as $\sqrt{\frac{2}{x-1}}$; neither survived the extraction from the PDF. The paper settles it twice over: the printed inverse is $\sqrt{x^2 + 1}$, and part (c) of the same question gives the volume of revolution about the y -axis as $\pi \left(\frac{h^3}{3} + h \right)$, which is exactly $\pi \int_0^h (x^2 + 1) \, dx$.

Solution 18 — justify existence, then find it

(a) On $[0, K)$,

$$g'(x) = -\frac{4x}{(1+2x^2)^2} < 0 \quad \text{for } x > 0,$$

so g is strictly decreasing: it has no turning point, it takes each value once, it is one-to-one, and therefore g^{-1} exists.

(b) Interchange and solve:

$$\begin{aligned}x &= \frac{1}{1+2y^2} \\ \Rightarrow 1+2y^2 &= \frac{1}{x} \\ &= \frac{1}{x} \\ \Rightarrow y^2 &= \frac{1-x}{2x} \\ &= \frac{1-x}{2x} \\ \Rightarrow g^{-1}(x) &= \sqrt{\frac{1-x}{2x}},\end{aligned}$$

the positive root because the values of g^{-1} lie in $[0, K)$.

For the domain, find the range of g : at $x = 0$, $g = 1$ (attained), and as $x \rightarrow K^- =$

$\frac{1}{\sqrt{2}}$ we get $2x^2 \rightarrow 1$ and $g \rightarrow \frac{1}{2}$, never reaching it. So the range of g is $(\frac{1}{2}, 1]$ and

$$\text{domain}(g^{-1}) = \frac{1}{2} < x \leq 1.$$

```
myl8a = sqrt((1 - x)/(2*x))  
myl8b = Interval.Lopen(Rational(1, 2), 1)
```

Marks. (a) A1 for " g is one-to-one", R1 for the reason — strictly decreasing, or no points of zero gradient. (b) M1 for rearranging and swapping, A1 for $y^2 = \frac{1-x}{2x}$, A1 for the positive root, A1 for the domain.

Two marks for a sentence. Part (a) has no algebra in it at all. The A1 is the claim, the R1 is the reason, and a candidate who jumps straight to the formula in (b) loses both.

The endpoints come from opposite ends. $x = 1$ is included because $g(0) = 1$ is attained; $x = \frac{1}{2}$ is excluded because K itself is not in the domain of g . One closed, one open, and each traceable to a specific end of the original interval.

Where this g came from. It is the sum of the series f_n of task 9, and $K = \frac{1}{\sqrt{2}}$ is the radius of convergence. The same question walks from a family of polynomials to a rational function to its inverse — which is the reason Paper 2 question 12 is worth 20 marks.

Solution 19 — completing the square inside an inverse

Interchange x and y in g and complete the square:

$$\begin{aligned}
 x &= \frac{1}{y^2 - 2y - 3} \\
 &\Rightarrow y^2 - 2y - 3 \\
 &= \frac{1}{x} \\
 &\Rightarrow (y - 1)^2 - 4 \\
 &= \frac{1}{x} \\
 &\Rightarrow (y - 1)^2 \\
 &= \frac{1 + 4x}{x}.
 \end{aligned}$$

Take the square root — both branches, then choose:

$$y - 1 = \pm \sqrt{\frac{4x + 1}{x}}.$$

The domain of g is $x > 3$, so the **values** of g^{-1} must exceed 3; the negative root gives values below 1 and is rejected. Hence

$$g^{-1}(x) = 1 + \sqrt{\frac{4x + 1}{x}} = 1 + \frac{\sqrt{4x^2 + x}}{x},$$

the last step being legal because $x > 0$.

For the domain: on $x > 3$ the expression $x^2 - 2x - 3$ increases from 0 (at $x = 3$, excluded) to ∞ , so g decreases from $+\infty$ towards 0, attaining neither. The range of g is $(0, \infty)$, so

$$\text{domain}(g^{-1}) = x > 0.$$

```
my19a = 1 + sqrt(4*x**2 + x)/x
my19b = Interval.open(0, oo)
```

Marks. M1 for interchanging, M1 for completing the square (or the quadratic formula), A1 for $(y - 1)^2 - 4$, A1 for $(y - 1)^2 = \frac{1}{x} + 4$, A1 for the $\pm$ line, R1 for rejecting the negative branch using $x > 3$; then A1 for the domain.

The R1 is a whole mark for one sentence. Not "take the positive root" — *why*: because g was defined only for $x > 3$, so g^{-1} can only produce values greater than 3.

Why $\sqrt{\frac{4x+1}{x}} = \frac{\sqrt{4x^2+x}}{x}$. Multiply inside by $\frac{x}{x}$: $\sqrt{\frac{4x^2+x}{x^2}} = \frac{\sqrt{4x^2+x}}{|x|}$, and $|x| = x$ because the domain is $x > 0$. On a domain containing negative numbers this step would be false.

A note on the archive. The corpus records the answer as $\frac{1+\sqrt{x^2+4x}}{x}$ — the bar and the root read one character too far. Test it: $g(4) = \frac{1}{5}$, so $g^{-1}(\frac{1}{5})$ must be 4. The paper's formula gives 4; the corpus's gives about 9.58. The check in the cell above rejects the corpus form on its own, without being told what the right answer is — which is the whole reason an inverse is verified by undoing rather than by comparison.

Solution 20 — an inverse through arcsin

Interchange, then apply $\sin$ to both sides:

$$\begin{aligned}x &= \arcsin \frac{y^2 - 1}{y^2 + 1} \\&\Rightarrow \sin x \\&= \frac{y^2 - 1}{y^2 + 1} \\&\Rightarrow \sin x (y^2 + 1) \\&= y^2 - 1.\end{aligned}$$

Collect the y^2 terms:

$$y^2 (\sin x - 1) = -1 - \sin x \Rightarrow y^2 = \frac{1 + \sin x}{1 - \sin x}.$$

The domain of g is $x \geq 0$, so the values of g^{-1} are non-negative, and the positive root is taken:

$$g^{-1}(x) = \sqrt{\frac{1 + \sin x}{1 - \sin x}}.$$

For the domain, find the range of g : at $x = 0$ the fraction is -1 , so $g(0) = -\frac{\pi}{2}$, attained; as $x \rightarrow \infty$ the fraction tends to 1 from below, so $g \rightarrow \frac{\pi}{2}$ without reaching it. Hence

$$\text{domain}(g^{-1}) = -\frac{\pi}{2} \leq x < \frac{\pi}{2}.$$

```
my20a = sqrt((1 + sin(x))/(1 - sin(x)))
my20b = Interval.Ropen(-pi/2, pi/2)
```

Marks. M1 for $x = \arcsin \frac{y^2-1}{y^2+1}$, A1 for applying sin and clearing, A1 for $y^2 = \frac{1+\sin x}{1-\sin x}$, R1 for taking the positive root with a reason, A1 for the expression; then A1 for the domain. Interval notations $[-\frac{\pi}{2}, \frac{\pi}{2})$ and $[-1.57, 1.57)$ are accepted.

The excluded endpoint is visible in the formula. At $x = \frac{\pi}{2}$ the denominator $1 - \sin x$ is zero. The domain and the algebra agree, which is a good sign that both are right.

Why f had to be cut down to g . The f of task 8 is even, so it is two-to-one and has no inverse; restricting to $x \geq 0$ throws away the mirror half. That restriction is what makes the positive root the right one — parity and branch choice are the same fact seen from two sides.

A note on the archive. The corpus records the function upside down, as $\arcsin \frac{1-x^2}{1+x^2}$. With that version the horizontal asymptote would be $y = -\frac{\pi}{2}$ and f would be increasing for $x < 0$ — both contradicting what the paper prints in the parts before this one. The markscheme settles it a third time: it gives $y^2 = \frac{1+\sin x}{1-\sin x}$, which is what the paper's version produces and the corpus's version does not.

Solution 21 — the timed task

(b)(iii) $m = 1$. The closed form is useless here — it reads $\frac{0}{0}$. Go back to the sum it came from:

$$f^n(x) = 1^n x + c(1 + 1 + \cdots + 1) = x + nc.$$

Or directly: $f(x) = x + c$, so $f^2(x) = x + 2c$, and each composition adds another c .

(d) $-1 < m < 1$. Then $m^n \rightarrow 0$, so

$$f^n(x) = m^n x + c \frac{1 - m^n}{1 - m} \rightarrow 0 + \frac{c}{1 - m}.$$

The limit does not depend on x : every point of every graph is pulled to the same height, so the family approaches the horizontal line

$$L: \quad y = \frac{c}{1-m}.$$

(e)(i) $m = -1$, n **odd**.

$$\begin{aligned} f^n(x) &= (-1)^n x + c \\ \frac{1 - (-1)^n}{1 - (-1)} &= (-1)^n x + c \\ &\quad \frac{1 - (-1)^n}{2}. \end{aligned}$$

For odd n , $(-1)^n = -1$, so $f^n(x) = -x + c \cdot \frac{2}{2} = -x + c$, as required.

(e)(ii) n **even**. Then $(-1)^n = 1$, so $f^n(x) = x + c \cdot \frac{0}{2} = x$.

```
my21a = x + n*c
my21b = c/(1 - m)
my21c = -x + c
my21d = x
```

Marks. (b)(iii): M1 for substituting $m = 1$ into their f^n , A1 for $x + c(1 + 1 + \cdots + 1)$, A1 for $x + nc$ — or the same three by iterating $f(x) = x + c$ directly. (d): M1 for $m^n \rightarrow 0$, A1A1 for the limit, A1 for writing

it as the equation of a line. (e)(i): A1 for the substituted form, R1 for $(-1)^n = -1$ when n is odd — then AG. (e)(ii): M1 for evidence that an even n was considered, A1 for $f^n(x) = x$.

L is the fixed point. Solving $f(x) = x$ gives $mx + c = x$, that is $x = \frac{c}{1-m}$ — the same number. Iterating a contraction pulls every starting point to the fixed point, which is what the picture of flattening graphs is showing.

Answer with an equation, not a number. Part (d) asks for the equation of L . " $\frac{c}{1-m}$ " alone is not a line; " $y = \frac{c}{1-m}$ " is, and the last A1 is for exactly that.

And here the practicum closes its loop. For $m = -1$ and even n , f^n is the identity — in particular $f \circ f = \text{id}$, which is the definition of $f = f^{-1}$. Check it directly: $f(x) = -x + c$ and

$$f(f(x)) = -(-x + c) + c = x.$$

So $y = -x + c$ is a self-inverse function, and it must be symmetric in the line $y = x$

— which it is, being perpendicular to it. Task 13 found another self-inverse function by algebra; the two questions are the same question, one asked in Paper 1 and one inside a Paper 3 investigation.

On the clock. Eleven marks in about seventeen minutes, and the substitutions are short. The time goes into (d): reading "the family of graphs approaches a straight line" as "take the limit of $f^n(x)$ and notice that x disappears" is the step being tested, and it is worth sitting still for thirty seconds before writing.

What was left out. Part (c) of this investigation proves the formula for f^n by induction, for eight marks — that belongs to A7, and part (d) is a limit, which is why it sits slightly outside this practicum's ladder.